

Karthik Viswanathan

✉ vkarthik095@gmail.com • 🏠 Website • 🎓 Scholar • 🔗 LinkedIn • 🐙 GitHub

I am currently pursuing a PhD at the University of Amsterdam, focusing on **interpretability** of large language models through **geometric analysis** of their internal representations. My interdisciplinary background began with **competitive programming**, expanded through developing machine learning models as a **surveillance analyst** at Goldman Sachs, and was further enriched during my master's studies in **theoretical physics**. This blend of algorithmic thinking, machine learning experience, and theoretical physics provides me with a distinctive perspective for bridging computational and theoretical approaches to scientific research.

Education

University of Amsterdam

PhD in Physics, Expected: December 31, 2025

Thesis: From Language Models to Cosmic Structures: A Geometric Perspective

Advisor: Dr. Jan Pieter van der Schaar

Amsterdam, Netherlands

2021–2025

University of Amsterdam

MSc in Physics and Astronomy (Theoretical Physics), cum laude

Thesis: Exploring the Spectral Theory/Topological Strings duality

Advisor: Dr. Marcel Vonk

GPA: 8.7/10

Amsterdam, Netherlands

2019–2021

Indian Institute of Technology Madras

B.Tech in Engineering Physics

Thesis: Real Space Renormalization and Applications to Machine Learning

Advisor: Dr. Ashwin Joy

GPA: 8.29/10

Chennai, India

2013–2017

Research Experience

Probing Geometry of Next Token Prediction Using Cumulant Expansion of the Softmax Entropy

Karthik Viswanathan, Sang Eon Park

ICML 2025: Workshop on High-dimensional Learning Dynamics

Persistent Topological Features in Large Language Models

Yuri Gardinazzi*, **Karthik Viswanathan***, Giada Panerai, Alessio Ansuini, Alberto Cazzaniga, Matteo Biagetti

Proceedings of the 42nd International Conference on Machine Learning (ICML 2025)

** Equal contribution*

Geometry of Tokens in Internal Representations of Large Language Models

Karthik Viswanathan, Yuri Gardinazzi, Giada Panerai, Alberto Cazzaniga, Matteo Biagetti

Under review

Cosmology with persistent homology: A Fisher forecast

Jacky Yip, Matteo Biagetti, Alex Cole, **Karthik Viswanathan**, Gary Shiu

Journal of Cosmology and Astroparticle Physics (JCAP), Vol. 2024, Issue 09

Workshop: Interpretability in LLMs using Geometric and Statistical methods

Karthik Viswanathan, Jan Pieter van der Schaar

Organized a two-day workshop with around 40 participants from Amsterdam and AREA Science Park (Trieste).

Professional Experience

Goldman Sachs

Surveillance Analyst

Developed ML models for anomaly detection in financial data

Manager: Prof. Howard Karloff

Bangalore, India

2017–2019

Goldman Sachs

Summer Intern

Implemented fast Personalized PageRank in a MapReduce framework

Mentor: Dr. Koushik Balasubramanian

Bangalore, India

2016

Awards & Distinctions

2020: Sander Bais Prize for Academic Merit

Institute for Theoretical Physics Amsterdam

For exceptional academic performance in the master's program

2015: ACM ICPC World Finals

Represented India in the international collegiate programming contest

Marrakech, Morocco

2013: International Olympiad in Informatics Training Camp

Selected after ranking in top 25 in Indian National Olympiad in Informatics

Bangalore, India

Organization & Mentoring

AREA Science Park, Trieste

Master Student Supervisor

Student: Giada Panerai

Thesis: Zig-Zag Persistence in Neural Networks Representations

2024

University of Amsterdam

Master Student Mentor

Student: Sibilla Bouche

Thesis: The persistence of non-Gaussian features: a Neural Ratio Estimation approach

2023

University of Amsterdam

Master Student Supervisor

Student: Sliem el Ela

Thesis: From Primordials to Persistence: A Dual Exploration of Multiparameter Topology and Cosmic Origins

2023

University of Amsterdam

Journal Club Organizer

Cosmology Journal Club

2022

Technical Skills

Programming: PyTorch, TensorFlow, Hadoop, MapReduce, Mathematica

HPC Resources: LUMI consortium, Snellius supercluster

Teaching

University of Amsterdam

Teaching Assistant

Courses: Topological Data Analysis: a Physics Perspective, Advanced Cosmology, ML for Physics and Astronomy, QFT in Curved Spacetimes, General Relativity

MSc Physics Program

2020–2024

Research Visits

2024: AREA Science Park, Trieste, Italy

September 16 – October 4

2024: AREA Science Park, Trieste, Italy

February 15 – April 27

References

Dr. Jan Pieter van der Schaar

University of Amsterdam

j.p.vanderschaar@uva.nl

Dr. Matteo Biagetti

AREA Science Park, Trieste

matteo.biagetti@areasciencepark.it

Dr. Magnus Botnan

Vrije Universiteit, Amsterdam

M.B.Botnan@vu.nl

Dr. Alberto Cazzaniga

AREA Science Park, Trieste

alberto.cazzaniga@areasciencepark.it